

Course Title	Linear Algebra		
Course Code	SC 223	Credit Structure	3-1-0 -4 (L-T-P-C)
Category	Core	Semester	Autumn Semester (AY 25-26)
Program	B.Tech		
Prerequisites	NA		
Course Objectives/ Brief Course Description	This course provides a foundational introduction to linear algebra, a branch of mathematics central to many areas of science and engineering. Topics include systems of linear equations, matrices, determinants, vector spaces, linear transformations, eigenvalues and eigenvectors, and applications of linear algebra in various fields. The course emphasizes both theoretical understanding and problem-solving skills.		
Evaluation/ Grading Policy	Insem – 1: 25% Insem – 2: 25% Final Exam: 35% Assignment/Quiz: 15%		
Course Materials/ References	Lecture notes. Lecture notes/ class notes will be posted and made available for students after the lecture. Books • P. Selinger, <i>Matrix Theory and Linear Algebra</i> (www.mathstat.dal.ca/~selinger/linear-algebra/) • David C. Lay, Steven R. Lay and Judi J. McDonald. <i>Linear Algebra and Its Applications</i> • Kenneth Hoffman, Ray Kunze, <i>Linear Algebra</i> • Sheldon Axler, <i>Linear Algebra done right</i>		
Detailed Course Content	Systems of linear equations. Matrices and elementary row operations. Gauss-Jordan elimination. Pivot positions and pivot columns. Applications of systems of linear equations (e.g. <i>balancing chemical equations, traffic flow in a network of streets, Google PageRank, polynomial interpolation etc.</i>). Vectors and vector equations. Linear combinations. Span of a set of vectors. Linear independence. Matrix equations. The column space and the null space of a matrix. Linear transformations. Matrix-vector multiplication. The standard matrix of a linear transformation. Matrix algebra: addition, transpose, matrix multiplication. Properties of matrix algebra. Invertible matrices. Applications of matrix algebra (e.g. <i>the Hill cipher, error correcting codes etc.</i>). Determinants. Determinants and elementary row operations, <i>cofactor expansion, Cramer's rule</i> . Determinants and area and volume. General vector spaces. Subspaces. Linear transformations of general vector spaces. Basis of a vector space and coordinates relative to a basis. Change of basis. Dimension of a vector space. The rank theorem. The dot product. Length of a vector, distance between vectors, orthogonality of vectors. Pythagorean Theorem. Orthogonal and orthonormal bases. The Gram-Schmidt process. Orthogonal projections. Least square solutions of systems of linear equations. Application: fitting least square lines and curves. General inner product spaces. Eigenvalues and eigenvectors. The characteristic polynomial of a matrix. Matrix diagonalization. Orthogonal diagonalization and spectral decomposition of		

	symmetric matrices. Singular Value Decomposition. Applications of SVD (e.g. <i>image compression, data analysis etc.</i>).
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Course Outcome: At the end of the course the students will be able to:

CO1: Represent systems of linear equations in vector and matrix form and solve it.

CO2: Perform matrix-vector multiplication and understand how this operation defines a linear transformation between R^n and R^m

CO3: Add, multiply, and transpose matrices determine whether a given matrix is invertible and compute its inverse if it exists

CO4: Compute determinants of matrices both by cofactor expansion and by row reduction and use Cramer's rule to solve systems of equations and to compute inverses of matrices compute areas of parallelograms and volumes of parallelepipeds using determinants.

CO5: Vector space, subspace, dimension, rank theorem etc.

CO6: Recognize if a given function between vector spaces is a linear transformation understand the notions of the kernel and the image of a linear transformation and their relationship to the null space and the column space of a matrix

CO7: Compute the inner product of vectors in R^n determine if a set of vectors in R^n is orthogonal compute the projection of a vector onto a subspace orthogonalize a set of vectors using the Gram-Schmidt process - solve least squares problems

CO8: Compute the characteristic polynomial of a matrix, find eigenvalues and eigenvectors of the matrix, diagonalization of a matrix, compute orthogonal diagonalization of symmetric matrix, compute the Singular Value Decomposition of a matrix.

POs-COs Matrix:

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
x	x		x								x

APPENDIX: Detailed Course Content (Session-wise/ Module-wise)

Tentative Lecture Schedule

Sl. No.	Topics	Lectures
1	Systems of linear equations. Matrices and elementary row operations. Gauss-Jordan elimination. Pivot positions and pivot columns. Applications of systems of linear equations (e.g. <i>balancing chemical equations, traffic flow in a network of streets, Google PageRank, polynomial interpolation etc.</i>).	6
2	Vectors and vector equations. Linear combinations. Span of a set of vectors. Linear independence. Matrix equations. The column space and the null space of a matrix. Linear transformations. Matrix-vector multiplication. The standard matrix of a linear transformation.	8
3	Matrix algebra: addition, transpose, matrix multiplication. Properties of matrix algebra. Invertible matrices. Applications	4

	of matrix algebra (e.g. <i>the Hill cipher, error correcting codes</i> etc.).	
4	Determinants. Determinants and elementary row operations, <i>cofactor expansion, Cramer's rule</i> . Determinants and area and volume.	2
5	General vector spaces. Subspaces. Linear transformations of general vector spaces. Basis of a vector space and coordinates relative to a basis. Change of basis. Dimension of a vector space. The rank theorem.	4
6	The dot product. Length of a vector, distance between vectors, orthogonality of vectors. Pythagorean Theorem. Orthogonal and orthonormal bases. The Gram-Schmidt process.	4
7	Least square solutions of systems of linear equations. Application: fitting least square lines and curves. General inner product spaces.	5
8	Eigenvalues and eigenvectors. The characteristic polynomial of a matrix. Matrix diagonalization. Orthogonal diagonalization and spectral decomposition of symmetric matrices.	7
9	Singular Value Decomposition. Applications of SVD (e.g. <i>image compression, data analysis etc.</i>).	3